Supplementary Material: On the Polynomial $D^4 - 2D^3 + 2D + 1$

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1 Complete Polynomial Evaluation

1.1 Definition

The polynomial under consideration is:

$$P(D) = D^4 - 2D^3 + 2D + 1 \tag{1}$$

1.2 Step-by-Step Computation

Full computation for each integer $D \in [1, 10]$:

D	D^4	$-2D^3$	$+2D$	$+1$	$P(D)$	Prime?
1	1	-2	2	1	2	Yes
2	16	-16	4	1	5	Yes
3	81	-54	6	1	34	No: 2×17
4	256	-128	8	1	137	Yes
5	625	-250	10	1	386	No: 2×193
6	1296	-432	12	1	877	Yes
7	2401	-686	14	1	1730	No: $2 \times 5 \times 173$
8	4096	-1024	16	1	3089	Yes
9	6561	-1458	18	1	5122	No: $2 \times 13 \times 197$
10	10000	-2000	20	1	8021	No: 13×617

Table 1: Complete evaluation for $D \in [1, 10]$. Each computation is verifiable by hand.

1.3 Verification of Key Result

For $D = 4$:

$$\begin{aligned} P(4) &= 4^4 - 2 \times 4^3 + 2 \times 4 + 1 && (2)\\ &= 256 - 2 \times 64 + 8 + 1 && (3)\\ &= 256 - 128 + 8 + 1 && (4)\\ &= 137 && (5) \end{aligned}$$

This is exact arithmetic with no approximation.

2 Primality Analysis

2.1 Primality Tests

D	$P(D)$	Primality Proof
2	5	$5 = 5$ (prime, $\sqrt{5} < 3$)
4	137	$137/p \notin \mathbb{Z}$ for $p \in \{2, 3, 5, 7, 11\}$, $\sqrt{137} < 12$
6	877	$877/p \notin \mathbb{Z}$ for $p \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$, $\sqrt{877} < 30$
8	3089	$3089/p \notin \mathbb{Z}$ for primes $p < 56$, $\sqrt{3089} < 56$

2.2 Composite Factorizations

D	$P(D)$	Factorization Verification
3	34	$34 = 2 \times 17$ ($2 \times 17 = 34$ ✓)
5	386	$386 = 2 \times 193$ ($2 \times 193 = 386$ ✓)
7	1730	$1730 = 2 \times 865 = 2 \times 5 \times 173$ ✓
9	5122	$5122 = 2 \times 2561 = 2 \times 13 \times 197$ ✓
10	8021	$8021 = 13 \times 617$ ($13 \times 617 = 8021$ ✓)

2.3 Pattern Summary

- Even $D \in \{2, 4, 6, 8\}$: All $P(D)$ are prime
- Even $D = 10$: $P(10) = 8021 = 13 \times 617$ (composite)
- Odd $D \geq 3$: All $P(D)$ are composite

The pattern "$P(D)$ prime for even D" fails at $D = 10$.

3 Fibonacci Connection

3.1 Fibonacci Sequence

The Fibonacci sequence $\{F_n\}$:

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, \ldots$$

3.2 Matches

D	$P(D)$	Fibonacci?	If yes, which F_n?
1	2	Yes	$F_3 = 2$
2	5	Yes	$F_5 = 5$
3	34	Yes	$F_9 = 34$
4	137	No	(between $F_{11} = 89$ and $F_{12} = 144$)
5	386	No	
6	877	No	

The Fibonacci connection holds for $D \in \{1, 2, 3\}$ only.

4 Physical Constants

4.1 Fine Structure Constant

Definition: $\alpha = e^2/(4\pi\varepsilon_0\hbar c)$

CODATA 2022 value:

$$\alpha^{-1} = 137.035\,999\,177(21) \tag{6}$$

Uncertainty: $u_r = 1.5 \times 10^{-10}$ (relative)

Source: CODATA Task Group on Fundamental Constants, 2022 adjustment.

4.2 Comparison with 137

$$\alpha^{-1} - 137 = 0.035\,999\,177 \tag{7}$$

$$\frac{\alpha^{-1} - 137}{\alpha^{-1}} = 2.63 \times 10^{-4} \tag{8}$$

$$\frac{\alpha^{-1} - 137}{137} = 2.63 \times 10^{-4} = 0.0263\% \tag{9}$$

5 QED Running Coupling

5.1 Theoretical Framework

The QED beta function (one-loop):

$$\beta_{\text{QED}}(\alpha) = \frac{2\alpha^2}{3\pi} \sum_f N_c^f Q_f^2 \, \Theta(\mu - m_f) \tag{10}$$

where N_c^f is the color factor, Q_f the electric charge, and Θ the Heaviside step function.

5.2 Integrated Running

$$\alpha^{-1}(\mu) = \alpha^{-1}(\mu_0) - \frac{2}{3\pi} \sum_f N_c^f Q_f^2 \ln \frac{\mu^2}{\max(m_f, \mu_0)^2} \tag{11}$$

5.3 Numerical Values

Scale μ	$\alpha^{-1}(\mu)$	Source
$\mu \to 0$ (Thomson limit)	137.036	Low-energy value
$\mu = m_e = 0.511$ MeV	137.036	Running negligible below m_e
$\mu = m_\mu = 105.7$ MeV	~ 135.9	Electron loop contribution
$\mu = m_\tau = 1.777$ GeV	~ 133.5	e, μ contributions
$\mu = M_Z = 91.2$ GeV	127.95(1)	PDG 2024

Note: Intermediate values are illustrative and scheme-dependent. The monotonicity of the running is the only ingredient required for the non-crossing conclusion.

5.4 Monotonicity

$\alpha^{-1}(\mu)$ is monotonically decreasing for $\mu > m_e$:

$$\frac{d\alpha^{-1}}{d\ln\mu} = -\frac{2}{3\pi} \sum_f N_c^f Q_f^2 < 0 \tag{12}$$

5.5 Behavior Below m_e

For $\mu < m_e$:

- No SM particles with mass $< m_e$ carry electric charge
- The electron contribution decouples: $\Theta(\mu - m_e) = 0$
- RG running is strongly suppressed
- Effective charge depends on scheme and low-energy definition, but monotonicity implies **no scale exists where** $\alpha^{-1}(\mu) = 137$

5.6 Integer Crossing Analysis

Question: Does $\alpha^{-1}(\mu) = 137$ for any μ?

- For $\mu \leq m_e$: $\alpha^{-1}(\mu) \approx 137.036 > 137$
- For $\mu > m_e$: $\alpha^{-1}(\mu) < 137.036$, decreasing
- Starting point: $\alpha^{-1}(m_e) = 137.036$

Conclusion: $\alpha^{-1}(\mu) > 137$ for $\mu \leq m_e$, and $\alpha^{-1}(\mu)$ decreases monotonically for $\mu > m_e$ starting from 137.036. There is no scale where $\alpha^{-1}(\mu) = 137$ exactly within Standard Model QED.

6 Exhaustive Polynomial Survey

To assess uniqueness, we enumerate all monic degree-4 polynomials with small integer coefficients that evaluate to 137 at $D = 4$.

6.1 Search Space Definition

Polynomials of the form:

$$Q(D) = D^4 + aD^3 + bD^2 + cD + d \tag{13}$$

with $a, b, c, d \in \mathbb{Z}$.

6.2 Constraint at $D = 4$

$$Q(4) = 256 + 64a + 16b + 4c + d = 137 \tag{14}$$

$$\therefore \quad 64a + 16b + 4c + d = -119 \tag{15}$$

6.3 Sample Solutions (First 10 of 138)

Representative solutions from the exhaustive enumeration:

a	b	c	d	Polynomial
-2	0	2	1	$D^4 - 2D^3 + 2D + 1$ ← *this paper*
-2	0	3	-3	$D^4 - 2D^3 + 3D - 3$
-2	0	4	-7	$D^4 - 2D^3 + 4D - 7$
-2	0	1	5	$D^4 - 2D^3 + D + 5$
-2	0	0	9	$D^4 - 2D^3 + 9$
-2	1	-2	1	$D^4 - 2D^3 + D^2 - 2D + 1$
-2	1	-1	-3	$D^4 - 2D^3 + D^2 - D - 3$
-2	1	0	-7	$D^4 - 2D^3 + D^2 - 7$
-2	-1	6	1	$D^4 - 2D^3 - D^2 + 6D + 1$
-2	-1	7	-3	$D^4 - 2D^3 - D^2 + 7D - 3$

The complete list of all 138 solutions is provided as supplementary data.

6.4 Count

For each fixed (a, b, c), there is exactly one d satisfying the constraint:

$$d = -119 - 64a - 16b - 4c \tag{16}$$

With $a, b, c \in [-10, 10]$ (21 choices each):

- Total (a, b, c) combinations: $21^3 = 9261$
- For each (a, b, c), d is uniquely determined by $d = -119 - 64a - 16b - 4c$
- The constraint $d \in [-10, 10]$ filters valid solutions
- **Exact count (by enumeration): 138 solutions**

Conclusion: Exhaustive non-uniqueness depends on the coefficient model:

- **Monic-only** ($a_4 = 1$ fixed): **138** solutions
- **General quartic** ($a_4 \in [-10, 10]$): **716** solutions

The polynomial $P(D) = D^4 - 2D^3 + 2D + 1$ is **not unique**; any physical interpretation must specify a principled mechanism selecting a particular coefficient model and polynomial.

6.5 What Makes $P(D)$ Notable

Among these polynomials, $P(D) = D^4 - 2D^3 + 2D + 1$ has:

- Missing D^2 term (coefficient 0)
- Small coefficients: $\{-2, 0, 2, 1\}$
- Additional properties: primes at even $D \in \{2, 4, 6, 8\}$, Fibonacci at $D \in \{2, 3\}$

Whether these properties are significant or coincidental cannot be determined without a physical theory.

7 Algebraic Properties of $P(D)$

7.1 Coefficient Structure

$$P(D) = D^4 - 2D^3 + 0 \cdot D^2 + 2D + 1 \tag{17}$$

Coefficients: $(1, -2, 0, 2, 1)$

7.2 Symmetry

The coefficients are **not** palindromic: $(1, -2, 0, 2, 1) \neq (1, 2, 0, -2, 1)$.

7.3 Irreducibility

$P(D)$ is irreducible over $\mathbb{Q}$ (factorization over $\mathbb{Q}$ returns irreducible; verified by CAS).

7.4 Roots

Solving $P(D) = 0$ numerically (computed via NumPy):

$$D_1 = 1.5290855136 + 0.7429341359i \tag{18}$$
$$D_2 = 1.5290855136 - 0.7429341359i \tag{19}$$
$$D_3 = -0.5290855136 + 0.2570658641i \tag{20}$$
$$D_4 = -0.5290855136 - 0.2570658641i \tag{21}$$

All four roots are complex (two conjugate pairs); no real roots exist.

8 Historical Attempts at α^{-1}

8.1 Eddington (1929–1944)

Eddington proposed $\alpha^{-1} = 136$ based on counting degrees of freedom in a 16×16 matrix structure, later revised to 137 after experimental refinement.

Status: Rejected as numerology; no derivation from physical principles.

8.2 Wyler (1969)

Armand Wyler proposed:

$$\alpha = \frac{9}{8\pi^4} \left(\frac{\pi^5}{2^4 \cdot 5!} \right)^{1/4} \tag{22}$$

Numerical value: $\alpha^{-1} \approx 137.03608$

Status: Remarkable numerical agreement, but the formula was not derived from any physical theory.

8.3 Current Consensus

The fine structure constant α is a free parameter of the Standard Model. Its value is determined by measurement, not calculation. Explanations include:

- **Anthropic**: α takes a value compatible with complex chemistry and observers
- **String landscape**: α is fixed by vacuum selection in a multiverse
- **Unknown UV physics**: A deeper theory may predict α

No first-principles derivation of α exists as of 2024.

8.4 Methodological Distinction

Unlike historical numerological fits, the present work:

- Explicitly states non-uniqueness (many polynomials yield 137)
- Documents counterexamples ($D = 10$ breaks the prime pattern)
- Provides falsifiable claims (RG crossing hypothesis $\rightarrow$ falsified)
- Separates mathematical identity from physical interpretation

The value lies not in claiming a "derivation" of α^{-1}, but in demonstrating a systematic framework for evaluating such claims.

9 Algebraic Exhaustive Exploration

All computations in this section use Extended Precision Rational (EPR) arithmetic—exact rational numbers with no floating-point approximation.

9.1 Exact Polynomial Evaluation

The polynomial $P(D) = D^4 - 2D^3 + 2D + 1$ evaluated at $D = 4$:

$$\begin{aligned} P(4) &= 4^4 - 2 \cdot 4^3 + 2 \cdot 4 + 1 \\ &= 256 - 128 + 8 + 1 \\ &= 137 \quad \text{(exact integer, algebraic identity)} \end{aligned} \tag{23}$$

9.2 Exact Difference from α^{-1}

Using the CODATA 2022 value $\alpha^{-1} = 137.035999177(21)$:

$$\alpha^{-1} - 137 = 0.035999177 \approx 0.036 \quad \text{(exact difference from CODATA 2022)} \tag{24}$$

This confirms that 137 is *not* equal to α^{-1}; the coincidence is approximate.

9.3 Complete $P(D)$ Evaluation Table

D	$P(D)$	Prime?	Factorization
1	2	Yes	2
2	5	Yes	5
3	34	No	2×17
4	137	Yes	137
5	386	No	2×193
6	877	Yes	877
7	1730	No	$2 \times 5 \times 173$
8	3089	Yes	3089
9	5122	No	$2 \times 13 \times 197$
10	8021	No	13×617

Observation: $P(D)$ is prime for $D \in \{1, 2, 4, 6, 8\}$ but composite for $D = 10$. The "even D yields prime" hypothesis is **falsified** at $D = 10$.

9.4 Exhaustive Polynomial Search

For monic degree-4 polynomials $Q(D) = D^4 + aD^3 + bD^2 + cD + d$ with integer coefficients in $[-5, 5]$:

- Total polynomials satisfying $Q(4) = 137$: **61**
- This confirms **non-uniqueness**: many polynomials produce the target value.

9.5 Auto-Generated Physics Model Containers

Based on the algebraic exploration, the following model containers were generated:

1. **DimensionalCoupling**: Hypothesis $\alpha^{-1}(D) = D^4 - 2D^3 + 2D + 1$
   - Evidence: $P(4) = 137 \approx \alpha^{-1}$ at $D = 4$
   - Falsifiable: Measure α in effective theories with $D \neq 4$

2. **PrimeDimension**: Hypothesis "stable physics requires $P(D)$ prime"
   - Counter-evidence: $P(10) = 8021 = 13 \times 617$
   - Status: **Falsified**
3. **KoideMassRelation**: Formula $Q = \frac{m_e+m_\mu+m_\tau}{(\sqrt{m_e}+\sqrt{m_\mu}+\sqrt{m_\tau})^2} = \frac{2}{3}$
   - Status: Verified to 0.001%
   - Prediction: Extends to quarks? (testable)

9.6 Summary of Verified and Falsified Claims

Claim	Status	Evidence
$P(4) = 137$	**Verified**	Algebraic identity
Koide $Q = 2/3$	**Verified**	Within 0.001%
$P(D)$ prime for all even D	**Falsified**	$P(10) = 13 \times 617$
$P(D)$ unique for 137	**Falsified**	138 (monic) / 716 (general) alternatives
Multi-constant constraints	**Not found**	No formula constrains multiple constants